Do Now · retrieval
Character sets: ASCII & Unicode

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which data type should store a telephone number beginning with 0? Explain briefly.

Q2

Convert 41_{16} to denary and eight-bit binary.

Q3

Apply AND 00001111 to 10100101. Explain which bits the mask preserves.

Q4

In 6+4 two’s-complement floating-point format, subtract 010000 0010 from 011000 0010. Give a normalised result.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Characters need agreed codes — A character set defines characters and their assigned numeric codes. An encoding specifies how those codes are represented as bytes.

Core idea 2

Standard ASCII is limited — Seven bits provide $2^7 = 128$ codes for letters, digits, punctuation and control characters. — Eight-bit extensions can provide 256 values, but different extensions may map the extra codes differently.

Core idea 3

Unicode supports a much larger repertoire — Unicode assigns code points to characters from many writing systems and symbols. Its first 128 code points match ASCII.

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

Use the encoding stated in storage questions — Storage depends on the number of code points and the number of bytes used for each one. Do not assume every Unicode character uses the same width. — CODE:
'0' in ASCII = 48 = 00110000_2
integer 0 = 0

Vocabulary · use the correct technical terms

Character set — A collection of characters with assigned codes.

Code point — A numeric identifier assigned to a character in Unicode.

Encoding — Rules for representing character codes as bytes.

ASCII — A standard seven-bit character code with 128 possible values.

Independent skill check · apply the lesson

Compare standard ASCII with Unicode for a multilingual app. Then encode the character C as an 8-bit ASCII byte (A = 65).

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Encode the text “AB”

Follow the worked method

1. Start with the character codes — CODE:
A = 65
B = 66 — ASCII assigns a numeric code to each character.

2. Write the 7-bit ASCII values — CODE:
A = 1000001
B = 1000010

3. Add a leading zero — CODE:
A → 01000001
B → 01000010

AB → 01000001 01000010 — The leading zero lets each 7-bit ASCII code sit inside an 8-bit byte; it does not expand the standard ASCII repertoire.

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · solo guided practice

Work ALONE. Try first; read the clues one at a time if you get stuck.

Complete the steps below. Check your answer only on page 8.

YOUR TASK

Build the answer

CLUES · uncover one at a time

Clue 1: The ASCII code for character '0' is 48.

Clue 2: The code for character '7' is seven more than the code for '0'.

Clue 3: An integer with value 7 is not the same as the ASCII code for '7'.

BUILD YOUR ANSWER · in BLACK PEN

Step 1: Find the ASCII code for the digit character.

Step 2: Convert that code to an 8-bit byte.

Step 3: Convert the integer 7 and explain the difference.

CHECK BEFORE TURNING THE PAGE

Have you shown every step? The worked self-check is at the END on page 8.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Question 2 · ADAPTED EXAM COMPARISON · [4]
A school app must store English and Chinese messages. Explain why standard ASCII is unsuitable [1], why Unicode is suitable [1], give one similarity between them [1], and explain why the receiver must know the encoding [1].
Contextual adaptation of OCR June 2022 Q5(e)(ii–iii); classroom marks.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Question 5 · ORIGINAL ENCODING APPLICATION · [4]
A UTF-8 string contains 20 ASCII code points, three euro-sign code points and two emoji code points. The byte widths are 1, 3 and 4 respectively. Ignore metadata. Calculate bytes [2] and bits [1]. Explain why multiplying the number of code points by 2 bytes is not valid here. [1]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Standard ASCII provides 128 character codes.

Use key vocabulary from memory

1. Explain Character set in your own words.

2. Explain Code point in your own words.

3. Explain Encoding in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

END SELF-CHECK · only after you have attempted the work

PAGE 5 · Guided We Do solution:

Character '7' has code 55 = 00110111; integer 7 is 00000111 as an 8-bit unsigned value.

PAGE 3 · Independent skill check:

Standard ASCII has 128 codes (7-bit), unsuitable for Chinese characters; Unicode has a far wider repertoire. C = 67 = 01000011 in an 8-bit byte.

Correct any errors in GREEN PEN. Write one improvement.

YOU DO ANSWERS: rhsccs.org/lessons/h446/lesson-008/answers.html